Physics Is Already There:
Rethinking Physical Injection in Latent World Models

Anonymous submission

Abstract

A world model predicts how an environment evolves in response to actions by rolling an internal representation forward, and is expected to respect physical regularities such as how objects move and collide. We study JEPA-style latent world models and find a puzzle: trained with no physics supervision, such a model already recovers object position well from its representation and predicts one step almost exactly, yet its multi-step rollouts drift away from the very dynamics it encodes. The popular fix is to inject physical structure into the latent representation. We test this exhaustively—ten injection variants spanning hard and soft constraints on both physical quantities and physical laws, with labeled and label-free supervision—across three controlled physics domains and two realistic simulation benchmarks, and find injection fails in 29 of 30 cases; even supplying the *correct* physical constraint makes prediction worse. We trace this to a single mechanism: the physical state is already encoded across the model's internal representation, so an injected copy is superfluous—prediction reads the state from the internal representation and bypasses the injected one. What the model lacks is not the object's physical state but the ability to evolve that state over long horizons: fixing the training procedure alone, adding no physics, cuts rollout error 2.2–8.3× on identical code and metrics. To help a latent world model, injected structure should supply what the representation is missing—*how the state evolves, not what the state is*.

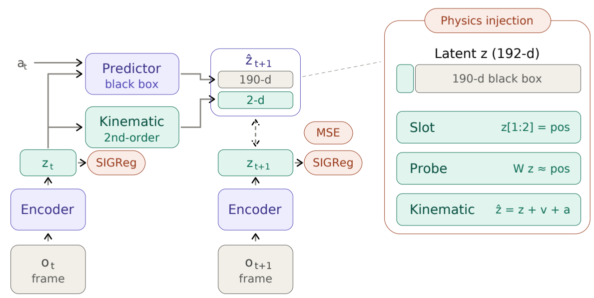

Figure 1: Physics injection into the LeWM shared-latent world model. The encoder maps each frame to a 192-d latent; a black-box transformer predicts the full latent from (z_t, a_t), and the free-rollout objective aligns the predicted $\hat{z}_{t+1}$ to the encoded next latent by MSE (SIGReg prevents collapse). Physics enters a 2-d position *slot* carved from the 192 dimensions (right panel): a hard *slot* pinning $z[1{:}2]$ to position, a supervised *probe* reading position off the latent, or a *kinematic* head evolving the slot by a second-order rule—while the other 190 "black-box" dimensions stay free. This 2-vs-190 split is the crux: prediction can route through the black box and bypass any injected slot.

1 Introduction

A *world model* rolls an internal representation forward to predict how an environment evolves under an agent's actions—a simulator to plan with instead of acting in the world (Ha and Schmidhuber 2018; LeCun 2022). For a physical environment, it should respect how objects move, fall, and collide. JEPA-style latent world models (Maes 2025; Balestriero and LeCun 2025) present a puzzle here. Trained on PhyWorld (Kang et al. 2024) with no physics supervision, such a model already recovers object position from its representation by linear probing (correlation up to 0.96) and predicts one step almost exactly (latent cosine 0.98–0.99)—yet rolled out autoregressively, its predictions drift away from the very dynamics it encodes: on collision the latent cosine falls to 0.24 by horizon 28, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold. The state is present in the representation; the model simply fails to *evolve* it.

The popular fix is to *inject physical structure into the latent* (Figure 1): a dedicated slot pinned to the physical state, a supervised probe that reads it out, or a hard-wired dynamics rule. Such injection has real successes in adjacent model classes—architectures that route prediction through a low-dimensional physical state by construction (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025), and deep supervision in a compact state latent where the supervised quantities occupy a large fraction of the representation (Zahorodnii 2025; Nie et al. 2026). Scaling data is not the answer either: video generators trained at scale still fail OOD physics, memorizing cases rather than laws (Kang et al. 2024). But for a *shared, high-dimensional* latent—where prediction runs end-to-end through one representation—injection has only been tried in isolated, incomparable settings, and never against the confound that dominates the outcome: the training protocol. Whether physical structure adds anything, once the protocol is done right, has never been tested systematically.

We answer this with a systematic, controlled anatomy on a compact JEPA world model. We cross ten injection variants—spanning hard and soft constraints, targeting the physical *state* (what it is) or its *evolution* (how it changes), under labeled and label-free supervision—with two training regimes and three controlled physics domains, and validate the surviving conclusions on two realistic-simulation benchmarks (Bear et al. 2021; Tung et al. 2023). Verdicts rest on scale-aware prediction error (latent nMSE and pixel PSNR), never on probe correlation or cosine—diagnostics that rise mechanically with the injected loss and can manufacture apparent gains; all headline comparisons are seed-controlled (§4.1).

The structure does not help. Injection degrades or fails to improve prediction in 29 of 30 mechanism–domain cells, and the failures are pointed. Supplying the *correct* physical form—the exact gravitational law $a{=}g$—hurts as much as an unconstrained dynamics head ($1.57\times$ the baseline error on uniform motion). Co-training the structure from scratch, rather than grafting it onto a pretrained encoder, hurts *more*, not less ($+0.558$ vs. $+0.035$ nMSE)—closing the defense that physics must be baked in during pretraining. The lone seed-robust gain reverses sign across domains: a diagnostic signature, not a usable prior.

Why should the *correct* physics hurt? Because the physical state is *already there*. A pinned slot occupies 2 of the 192 latent dimensions, but the other 190 redundantly encode the same position—decoding it nearly as accurately as the full latent—so prediction routes around the injected slot and reads the copy it already carries. This account makes a falsifiable prediction: if the slot's $\sim$1% share of the prediction loss is what starves it, up-weighting the slot should remove the harm. It does not—re-weighting to saturation recovers only *parity*, never a net gain, because no loss weight closes a bypass wired into the architecture. The state is *decodable, but not load-bearing*.

What the representation lacks is not the state but the ability to *evolve* it over long horizons—and that gap is fixable with no physics at all. Repairing the training protocol alone—free rollout, with the horizon matched to each domain's dynamics—cuts rollout error 2.2–8.3$\times$ on the identical code, metrics, and seeds that judged every injection cell, pinning the failure on the injected structure rather than the pipeline or the yardstick. The lesson is constructive: to help a shared-latent world model, an inductive bias must supply what the representation is missing—how the state evolves—not re-encode what it already holds. An architecture that makes the physical state the *sole* pathway prediction can take does close the bypass; but ported faithfully to our benchmark, it collapses once an OOD shift reaches its encoder (ρ 0.33 vs. 0.89)—necessary, but not sufficient. Every finding above replicates on a second, frozen-DINOv2 backbone.

This paper contributes:

- the first systematic, seed-controlled anatomy of physical inductive biases in latent world models—five mechanism families, ten variants, two training regimes, three synthetic domains, two realistic-simulation benchmarks—showing injection fails as a general lever (29/30; §4.2), replicated on a second, frozen-DINOv2 backbone (§4.5);
- a mechanistic account, *decodable but not load-bearing*, with three tests: a black-box bypass probe, a loss-dominance and effective-dimensionality measurement that rules out implementation failure, and a dose–response re-weighting sweep (§4.3);
- a controlled factorial showing the dominant variable is the training protocol, not physical structure—doubling as the positive control that excludes implementation and metric artifacts (§4.4);
- an architectural attribution: a faithful extrinsic port closes the bypass but still collapses under encoder-OOD shift, making extrinsic design necessary but not sufficient (§4.5).

2 Related Work

Physically structured world models. A growing line of work argues that world models should carry explicit physical state. Physically interpretable world models align a dedicated low-dimensional state with physical quantities and route prediction through it (Mao, Umasudhan, and Ruchkin 2024); design principles for such models are catalogued by Peper et al. (2025); deep supervision injects physical readouts into the training loss and improves prediction in a low-dimensional state latent, where the supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025); closest to our setting, Phys-JEPA decomposes a JEPA latent into physical and residual components and supervises the physical component against observed variables, reporting gains (Nie et al. 2026)—but on *numerical multivariate time series* (low-dimensional tabular data, no video), not the visual world models we study. The gap is central: a supervised component that is load-bearing in a low-dimensional numerical latent becomes a small (2/192), redundantly encoded fraction that prediction bypasses in a visual latent (§4.3), reversing the conclusion. Hamiltonian networks impose conservation structure on the dynamics (Greydanus, Dzamba, and Yosinski 2019). We acknowledge these successes in their own settings, and test the transferable core of the proposals—fixed slots, kinematic equations including the exact gravitational form, probe supervision, consistency losses, and label-free variants—inside a shared-latent *visual* JEPA world model, where we find all of them unhelpful or harmful (§4.2). Our external port of the extrinsic architecture (§4.5) attributes the discrepancy to architecture rather than to the physics.

Latent world models and physics benchmarks. JEPA-style models predict in representation space (Assran et al. 2023; Bardes et al. 2024; Balestriero and LeCun 2025; Maes 2025); DINO-WM plans over frozen pre-trained visual features (Zhou et al. 2025; Oquab et al. 2024), a design we adopt as our cross-backbone control (§3.4); Dreamer-line models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2023). PhyWorld (Kang et al. 2024) generates controlled single-mechanism physics videos with explicit in-distribution ranges and showed that scaled video-diffusion generators memorize cases rather than laws; Physion and Physion++ (Bear et al. 2021; Tung et al. 2023) provide realistic—not camera-captured—simulation with rich multi-object dynamics. We use PhyWorld's OOD protocol to dissect *latent* world models rather than pixel-space generators, and validate surviving conclusions on Physion/Physion++.

Rollout training and exposure bias. Training autoregressive predictors on their own outputs is a classic remedy for compounding error: scheduled sampling (Bengio et al. 2015), Data-as-Demonstrator (Venkatraman, Hebert, and Bagnell 2015), and professor forcing (Goyal et al. 2016) all attack the train–test mismatch of teacher forcing. We claim no novelty for free rollout. Its role in this paper is methodological (§3.3): a controlled factorial variable—the confound prior injection studies never controlled—and a positive control that separates "physical structure does not help" from "our implementation or metrics are broken."

3 Method: A Design Space of Physical Injection

We answer the question of §1 by construction: design the ways physics can be injected into a shared latent, state what each design is supposed to achieve, and lay out the experiments that test the mechanism behind it.

3.1 Injection Design Space: Five Families, Ten Variants

We span the design space along three orthogonal axes—*hard vs. soft* constraints on the physical quantity, pinning *what the motion state is* vs. *how it evolves*, and *labeled vs. label-free* supervision—so that any "try a different encoding" objection lands on an axis we have measured. Table 1 lists the ten variants with the design rationale of each.

Orthogonally, we cross injection with the *training regime*: fine-tuning a pretrained encoder vs. from-scratch co-training; with injection on/off this gives a 2×2 per domain. This closes the remaining escape hatch—"structure must be baked in during pretraining, as extrinsic architectures do."

Variant	What it does, and what it tests
Family 1: fixed slot — hard constraint, pins the state, labeled	
slot (structpos)	pin dims $[0{:}2]$ to the true position (the object's x, y coordinates): the most direct state injection; tests whether a hard constraint helps
+reweight	raise the slot's share of the prediction loss (weight 30): if failure is due to the slot's $\sim 1\%$ gradient share, this should rescue it
+velocity	encode velocity too: tests whether it matters *which* quantity is injected, not just that something is
Family 2: deep-supervision probe — soft constraint, pins the state, labeled	
probe	linear head reads position from the latent: replicates the deep-supervision recipe (Zahorodnii 2025) in a high-dimensional visual latent
+slot	soft + hard combination: tests complementarity
Family 3: kinematic head — hard constraint, pins the evolution, labeled	
free MLP	attach $z{+}v{+}a$ update with learned a: upgrade from pinning state to pinning *evolution*
strict $a{=}g$	exact gravitational form, one learnable constant: closes the "your physics was wrong" objection
Family 4: consistency — soft constraint, pins the evolution, labeled	
consistency	finite-difference velocity of the rolled-out positions must match ground truth; assumes no form for a, designed for collision impulses
Family 5: label-free prior — soft constraint, pins the evolution, label-free	
label-free	slot must evolve as smooth second-order dynamics, no ground truth: the only injection available for raw video
grounded	same structure + true labels: isolates the label variable

Table 1: Ten injection variants in five families, spanning three axes (read off each family's header). **Constraint**—*hard*: a latent dimension is forced to equal the physical quantity; *soft*: an auxiliary loss only encourages it. **Target**—the *state* (what the motion is: position, velocity) vs. its *evolution* (how it changes: a dynamics rule). **Supervision**—*labeled* uses ground-truth physics, *label-free* does not. The three axes span the ways to inject physics into a shared latent, so any alternative encoding falls on an axis we test. All variants use the strongest training protocol (§3.3).

3.2 Mechanistic Hypothesis and Its Tests

Hypothesis (the load-bearing problem). The physical slot occupies 2 of 192 dimensions and receives $\sim 1\%$ of the prediction gradient, while the black-box channel *redundantly* encodes the same position—so prediction can route around any injected slot. Because the latent already carries the injected state, injection adds no new signal—it only consumes capacity and gradient share. We test this account in three ways, each admitting a specific outcome that would falsify it:

1. **Is the physical constraint injected?** Split the latent into the 2 slot dimensions and the 190 black-box dimensions and decode position from each separately, with a random-2-dimension control. If the black box alone cannot decode position, the redundancy claim is false.
2. **Is the constraint acting at all?** Two measurements: how large the weighted physical auxiliary loss is relative to the prediction loss (a proxy for gradient magnitude), and whether the encoder's effective dimensionality (participation ratio) changes. If neither moves, the constraint never took effect and the failure is a bug. If the auxiliary term dominates and dimensionality collapses, the constraint is acting—and competing with prediction.
3. **Does re-weighting the physical loss help?** Sweep the

physical slot's prediction-loss weight from 1 to 300. If harm does not respond systematically to the weight, the gradient-share mechanism is wrong; if it responds but weighting to saturation still cannot rescue every domain, demonstrating that insufficient physical loss is not the root cause.

3.3 The Training-Protocol Control: Free Rollout

Free rollout serves two methodological roles; neither is a novelty claim (it is scheduled sampling, Bengio et al. 2015).

Controlled factorial variable. Physical structure has never been compared against the training protocol on the same baseline; prior studies treat the protocol as background. We set the protocol first—replacing teacher forcing (one-step, ground-truth context) with free rollout ($H{=}8$ autoregressive steps, supervising the whole rollout)—and then ask whether structure adds incremental value. Teacher forcing hides compounding error at training time (*exposure bias*): the model never sees its own mistakes, then must consume them at deployment. Free rollout exposes them during training, injecting no physics whatsoever. We read the resulting gain as more than robustness to compounding error: supervising an H-step autoregressive rollout turns training itself into *long-horizon dynamics simulation*. Under teacher forcing the predictor is graded only on reproducing local one-step transitions; under free rollout it is graded on *evolving* the state over the full horizon, so the gradient rewards a self-consistent multi-step dynamics rather than a one-step map—precisely the capability the latent is missing (§1). On this reading, free rollout is an evolution-targeted training signal that requires no physics labels: it trains *how the state evolves* without touching *what the state is*. A second protocol variable, the rollout horizon H, must *match dynamics complexity*: domains with late causal events (collision; realistic-simulation scenes) need horizons long enough to contain them.

Positive control. A negative result invites two deflations—"your implementation is broken" or "your metrics cannot detect improvement." A protocol change that produces large gains on identical code and identical metrics rules out both at once, pinning the failure on the injected structure itself.

3.4 External and Cross-Backbone Controls

Two controls probe our account from outside the LeWM backbone: one tests whether an architecture that *closes* the bypass succeeds where injection fails, the other whether the conclusions survive a different encoder.

Extrinsic port. We faithfully port the physically interpretable world model (PIWM) of Mao, Umasudhan, and Ruchkin (2024)—a three-stage pipeline (VAE encoder, 128-D; a supervised extractor to a low-dimensional physical state; known-equation dynamics with learnable constants)—onto the same PhyWorld domains and OOD protocol. The port tests our mechanism's architectural prediction from the outside: extrinsic designs make the physical state the sole pathway prediction must route through, closing the bypass by construction.

Cross-backbone control. To test that the conclusions are not artifacts of one encoder, we replicate the core protocol on a second JEPA-family instantiation in the style of DINO-WM (Zhou et al. 2025): a *frozen* DINOv2-small encoder (Oquab et al. 2024)—generic self-supervised features that have never seen PhyWorld—with the trainable projector serving as adapter and the identical predictor stack, losses, seeds, and evaluation. The variant differs from LeWM in backbone provenance, dimensionality, and, crucially, in whether the injected losses can reshape the encoder at all.

4 Experiments

4.1 Setup

Model. LeWM (Maes 2025), a compact JEPA-style action-conditioned world model ($\sim$10M parameters): a ViT-tiny encoder producing a 192-dimensional embedding, an MLP projector, an action encoder, and a 6-layer causal-transformer predictor, trained to minimize latent prediction error plus a SIGReg anti-collapse regularizer (Balestriero and LeCun 2025). Unless stated otherwise, the encoder is initialized from the public PushT checkpoint; §4.2 additionally studies from-scratch training. The compact, fully controlled backbone allows the complete factorial anatomy ($\sim$200 training runs) with headline configurations repeated over three seeds (3072/1234/42); remaining cells are single-seed and marked as such.

Datasets and OOD protocol. All benchmarks are *visual* world-modeling tasks (RGB video frames), not the numerical/tabular sequences of prior physics-injection work (§2); injected state is thus always a small fraction of a high-dimensional visual latent. *PhyWorld* (Kang et al. 2024) provides three single-mechanism synthetic domains—*uniform motion*, *parabola*, *collision* (zero/constant/impulsive acceleration)—each with 1,000 ID trajectories (32 frames, 224×224; radius/mass $\in [0.7, 1.5]$, velocity $\in [1, 4]$), evaluated on four partitions (ID, r/m-OOD, v-OOD, both-OOD); its single-variable OOD ranges isolate physical extrapolation from data-scale confounds. Two *realistic-simulation* benchmarks (rendered, not camera video) add realism: *Physion* (Bear et al. 2021) for zero-shot object-contact-prediction (OCP) transfer, and *Physion++* (Tung et al. 2023)—ten physical-property scenarios (mass/friction/bounce/deformation)—for direct training with rollout to horizon 64.

Metrics and decision rules. Verdicts use two scale-aware, *external* metrics: latent nMSE ($\|\hat{\mathbf{z}}-\mathbf{z}\|^2$ over target variance) and pixel PSNR of decoded rollouts—external because no injection variant optimizes them directly, and they agree in every case we examined. Latent cosine and probe decodability ρ serve as *diagnostics only*: both are duals of the auxiliary training losses and rise mechanically when those losses are optimized, so as verdicts they systematically overstate physical competence (we measured reversals up to cosine 1.50× better while nMSE degrades to 0.21×; our own early sweeps were misled this way before re-judging on prediction loss). nMSE has the opposite failure mode: its denominator degenerates when target variance collapses (a parabola ball exiting the frame drives late-horizon nMSE to 10^6 while cosine stays normal), so parabola is judged on its clean r/m-OOD

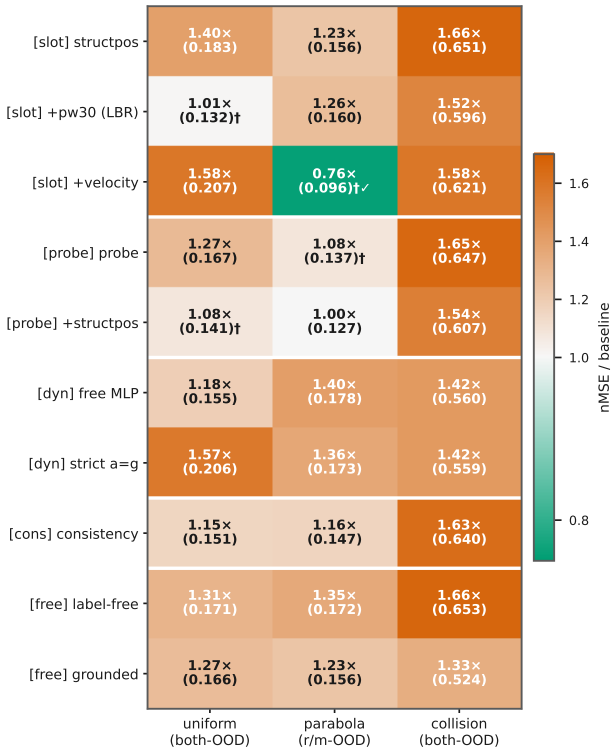

Figure 2: Every injection variant × domain against the free-rollout baseline (nMSE ratio; number = single-seed value, † = three-seed mean). Red = worse, white = parity, green = better. Injection is at or below baseline in 29 of 30 cells (25 worse, 4 parity); the collision column (1.33–1.66×) is uniformly worst; the sole real gain is velocity+slot on parabola (0.76×, †✓).

partition and all numbers are reported per partition and per horizon. Full pathology catalogue and protocol checklist: Appendix E. Splits are trajectory-level; three-seed results report mean±sample std.

4.2 Injection Fails in 29 of 30 Cells

Figure 2—whose cells carry the raw nMSE alongside the ratio, so it doubles as the full result table (reproduced with seed annotations in Appendix B)—answers every design rationale of Table 1, family by family. *Hard state injection*: the slot faithfully absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$; slot decodability rises $0.31 \rightarrow 0.96$), yet prediction worsens in all three domains ($1.40\times/1.23\times/1.66\times$)—the constraint acts, harmfully. *Soft injection*: the deep-supervision recipe, effective in a low-dimensional state latent where supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025), does not transfer to a 192-D visual latent where they occupy 2%: all three domains degrade ($1.27\times$/parity-after-seeds/$1.65\times$), and combining soft with hard does not rescue ($1.54\times$ on collision). *Correct physics does not help*: the strict $a{=}g$ head—the ground-truth functional form—degrades uniform by $1.57\times$ and parabola from 0.127 to 0.173, no better than an unconstrained MLP (0.178). *The purpose-built design fails worst on its target*: the consistency loss, designed for collision impulses, is among the worst exactly there ($1.63\times$). *Labels are not the issue*: the label-free prior and its perfectly labeled counterpart fail in the same direction (collision $1.66\times$ vs. $1.33\times$)—the failure is architectural, not informational.

The sole exception is the mechanism's signature, not a usable prior: adding *velocity* to the re-weighted slot helps on parabola (r/m-OOD $0.122{\pm}0.007 \rightarrow 0.096{\pm}0.007$, every seed lower) because there velocity is an extrapolable driving variable ($a{=}g$ makes it linear in time); the same encoding is the *worst* variant on uniform (+0.076, velocity is a redundant constant) and clearly harmful on collision (+0.025 over the reweighted slot; velocity jumps discontinuously). A prior that requires knowing the domain's dynamics in advance, and reverses sign across domains, is a diagnostic, not a method—and its best case (-0.026 nMSE) is an order of magnitude below the protocol lever of §4.4.

Domain	Regime	phys. off	phys. on	Δ
uniform	scratch	0.192	0.750	**+0.558**
	fine-tune	0.131	0.166	+0.035
parabola (r/m-OOD)	scratch	0.343	0.375	+0.03
	fine-tune	0.124	0.198	+0.07
collision	scratch	0.538	0.635	+0.097
	fine-tune	0.393	0.525	+0.132

Table 2: Injection on/off × training regime, per domain (latent nMSE ↓ on the hardest clean split; injected structure: strict $a{=}g$ head + fixed slot). *Scratch* trains everything from random initialization; *fine-tune* starts the encoder from pretrained weights. Δ is the cost of switching injection on within a regime: positive in all six regime–domain cells, and amplified from +0.035 to +0.558 on uniform when physics is baked in from the start.

From-scratch co-training hurts more. The remaining defense—structure must be baked in during pretraining—fails in the strongest way (Table 2): the physics penalty on uniform grows from +0.035 (fine-tuned) to +0.558 (from-scratch); all six regime–domain cells are positive. The cleaner the baseline, the larger the harm.

Two corroborations independent of PhyWorld nMSE. First, *zero-shot transfer* (different dataset, metric, and task): on Physion OCP, no synthetic-trained configuration exceeds the random-encoder prior (0.607 mean AUC); the physics-structured variant is the *worst* of all (0.551), below free rollout (0.603)—"more structure, worse transfer" (Appendix E). Second, *realistic simulation*: trained directly on Physion++, every injection variant we ported (fixed slot, consistency, consistency+acceleration) degrades per-scenario rollout nMSE by 3–10× against the identically trained free-rollout model (Table 3); the gaps dwarf the three-seed noise band of the baseline. The probe family has not yet been run on Physion++, so this corroboration currently covers the slot and consistency families.

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	**0.041**	0.119	0.078	0.067
mass_collision	**0.083**	0.510	0.682	0.696
mass_dominoes	**0.058**	0.452	0.586	0.597
mass_waterpush	**0.122**	0.363	0.663	1.476

Table 3: Physion++ direct training: per-scenario rollout nMSE (↓) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10× on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed the baseline's seed noise.

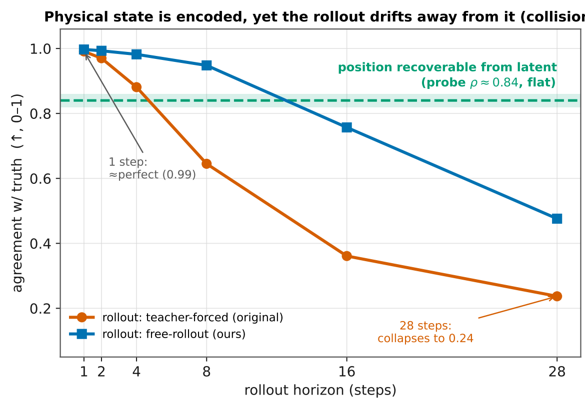

Figure 3: Decodable, but not load-bearing (collision domain). The flat dashed line is position decodability from the frame embeddings (probe $\rho \approx 0.84$): the state is *present* at every horizon. Yet the teacher-forced rollout (LeWM's default) collapses from near-perfect one-step agreement (0.99) to 0.24 by horizon 28; training the same model by free rollout (§3.3) stabilizes the long horizon (0.48) with no physics added. Presence of the state does not imply its use by prediction.

4.3 Mechanism: Decodable but Not Load-Bearing

The puzzle of §1 localizes here: position is linearly decodable from the latent at every horizon, yet the rollout drifts away from it (Figure 3)—*decodable, but not load-bearing*. The three tests explain why.

Test 1: the physical constraint is injected. Probing *only the* 190 *non-slot dimensions* decodes position at $\rho = 0.79$–0.92 across all three domains—as well as the full latent, undiminished by pinning the slot—while a random-2-dimension control fails (0.2–0.5). Position is a redundant, distributed copy the predictor can route through while ignoring any structured slot.

Test 2: the constraint acts harmfully. Both measurements move, which indicates the constraint acts. At probe weight λ=10 the weighted physical loss grows to 15–125× the prediction loss (the gradient-magnitude proxy), and the encoder's effective dimensionality collapses by 39–90%; at λ=1, the setting of §4.2, the ratio is milder (3–20×) yet the harm persists. With slot decodability rising $0.31 \to 0.96$, the constraint demonstrably reshapes the representation—in a direction that competes with prediction.

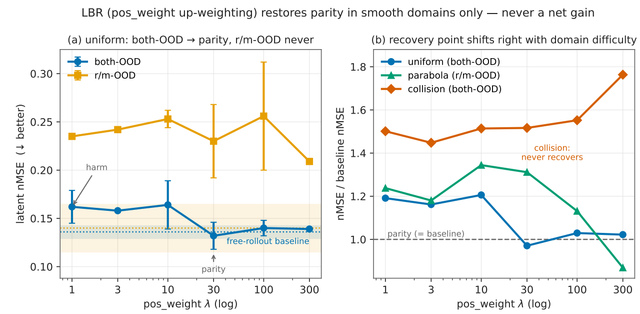

Figure 4: Dose–response re-weighting. Left: uniform, slot weight $1 \to 300$ against the baseline band—both-OOD returns to parity, r/m-OOD never does. Right: nMSE ratio to baseline across domains—collision stays above baseline at every weight and degrades as weight grows. Re-weighting validates the mechanism's direction but is not a repair.

Test 3: re-weighting the physical loss does not help. Sweeping the slot weight $1 \to 300$ (Figure 4) moves the harm systematically—the mechanism's direction is confirmed—but weighting rescues only 2 of 4 domain–partition decision cells, to *parity* and never a net gain: uniform both-OOD returns exactly to baseline (0.132 ± 0.017 vs. 0.136 ± 0.008), uniform r/m-OOD never recovers, and collision worsens monotonically with weight. Insufficient physical loss is therefore not the root cause: the black-box route stays open at every weight, and closing it requires changing the architecture.

4.4 Free Rollout Reduces Error by 2.2–8.3×

Replacing teacher forcing with free rollout cuts the hardest-split error 2.2× (uniform, $0.300 \pm 0.007 \to 0.136 \pm 0.008$), 3.6× (parabola, $0.443 \pm 0.048 \to 0.122 \pm 0.007$), and 2.4× (collision, $1.152 \pm 0.048 \to 0.479 \pm 0.079$), with improvements on *every* partition including ID (2.0–4.6×)—it is not an OOD patch (Figure 5). On Physion++ the same switch is worth 8.3× at horizon 64 ($1.174 \pm 0.041 \to 0.141 \pm 0.018$, three seeds), and the entire pattern replicates on the frozen-DINOv2 backbone (Figure 5, right; §4.5). The horizon must match dynamics complexity: collision improves markedly at $H \approx 20$ ($0.393 \to 0.294$) while smooth domains degrade with excess horizon (uniform horizon-28 cosine $0.969 \to 0.814$ at H=16); realistic-simulation scenes have the largest appetite—extending H=$8 \to 28$ (with geometric scale jitter stabilizing amplitude; Appendix D) drives horizon-64 nMSE from 0.280 to 0.014, a ∼19× reduction with no inflection found (Figure 6).

As the positive control of §3.3, this result carries the methodological weight of the paper: on the same code, metrics, seeds, and data that judged all thirty injection cells, a physics-free intervention produces multiples—so the injection failure of §4.2 is attributable to the injected structure, not to the pipeline or the yardstick.

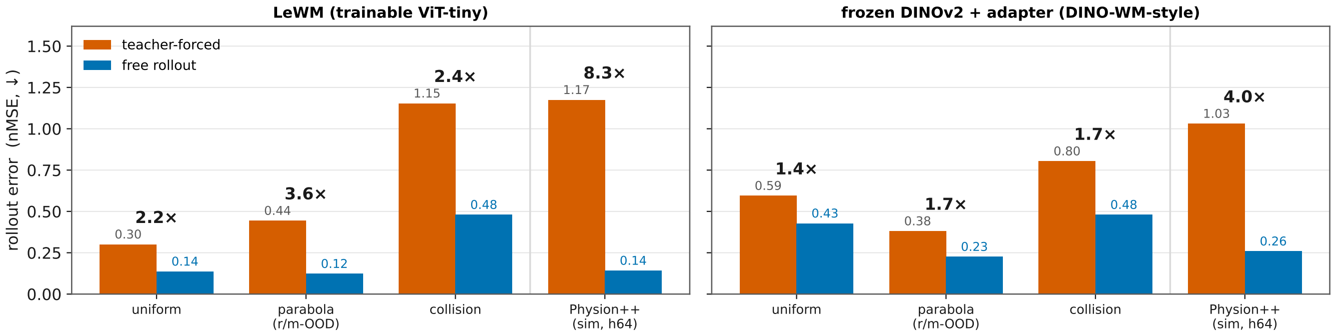

Figure 5: Teacher forcing vs. free rollout on both backbones (hardest clean split, nMSE ↓; fold-change above bars; all three-seed with non-overlapping intervals). Left, LeWM: uniform $0.300 \to 0.136$ (2.2×), parabola $0.443 \to 0.122$ (3.6×), collision $1.153 \to 0.479$ (2.4×), Physion++ horizon-64 $1.174 \to 0.141$ (8.3×). Right, the frozen-DINOv2 variant (§3.4): the same physics-free switch yields 1.4×/1.7×/1.7×/4.0×—the protocol lever is backbone-independent.

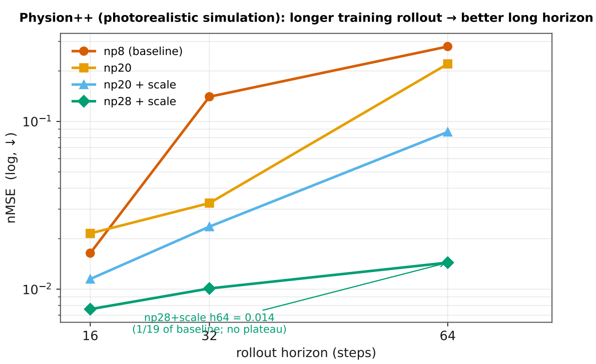

Figure 6: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H{=}8 \to 28$, +scale jitter). Long-horizon error falls monotonically to 1/19 of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

Partition (uniform)	Extrinsic port	LeWM (FR)
ID	**0.96**	0.93
r/m-OOD	0.33	**0.89**
v-OOD	**0.97**	0.87
both-OOD	0.48	**0.87**

Table 4: Rolled-out position decodability ρ (↑) of the faithfully ported extrinsic model vs. LeWM with free rollout. The extrinsic design wins where its encoder is in-distribution and collapses where the shift hits the encoder (radius/mass changes appearance): closing the bypass is necessary, not sufficient.

4.5 External Control: Extrinsic Is Necessary, Not Sufficient

Extrinsic port. The faithful extrinsic port (PIWM; §3.4) learns the correct physics and, where its encoder is in-distribution, beats our shared-latent model: rolled-out position ρ reaches 0.96 (ID) and 0.97 (v-OOD) vs. LeWM's 0.93/0.87 (Table 4). But under radius/mass shift—which changes appearance, hitting the VAE encoder—it collapses to $\rho = 0.33$ (r/m-OOD) and 0.48 (both-OOD) while LeWM holds 0.89/0.87. The architectural prediction of §4.3 is confirmed from the outside: making the physical state load-bearing by construction is what extrinsic designs buy (their equations alone do not transfer, §4.2); yet closing the bypass leaves the encoder-OOD problem unsolved. Extrinsic structure is necessary but not sufficient.

Cross-backbone replication. On the frozen-DINOv2 variant every headline finding replicates across three seeds. Presence is even stronger—the physics-naive frozen encoder decodes position at $\rho{=}0.95$ (vs. 0.90 for LeWM), with the same black-box bypass—so presence and bypass are properties of generic visual features, not our training. The protocol lever transfers (1.4–1.7× synthetic, 4.0× Physion++; Figure 5, right). Injection still never wins (27/30 at or below baseline; Appendix C); the three above-parity cells dissolve under content-free controls—a shuffled target, or a 10×-weaker pin at matched loss weighting, keeps the apparent gain while ID error degrades, i.e. capacity-restriction regularization, not physics (Trap 5, Appendix E). One difference: with the encoder frozen, injection is inert rather than harmful—the losses reshape only the adapter, bounding damage but foreclosing benefit.

4.6 Synthesis

Same latent, same code, same metrics: injecting state the latent already encodes fails in 29/30 cells (§4.2) because prediction routes around it (§4.3); repairing the long-horizon dynamics of the predictor—with no physics at all—yields 2.2–8.3× (§4.4); closing the bypass architecturally helps only until the encoder meets OOD physics; and the whole pattern survives a backbone swap (§4.5). Physical state in a latent world model is *decodable but not load-bearing*.

5 Conclusion

We asked whether injecting physics can make a latent world model physically consistent, and answered with a controlled anatomy: across five mechanism families, ten variants, two training regimes, three synthetic domains, and two realistic-simulation benchmarks, injection fails in 29 of 30 cells—with the exact gravitational form, with perfect labels, and *more* when co-trained from scratch. The reason is that the physics is already there: the shared latent redundantly encodes the state being injected ($\rho = 0.79$–0.92 from the non-slot dimensions alone), so injection adds no signal while consuming capacity and gradient, and prediction routes around it—*decodable but not load-bearing*. All three tests support this account, and its two predictions held: reweighting removes the slot's harm but only to parity, and an extrinsic architecture that closes the bypass by construction does help—until the OOD shift hits its encoder (ρ 0.33 vs. 0.89), making extrinsic design necessary but not sufficient. What actually moved prediction was never the structure: repairing the training protocol—free rollout, and horizons matched to dynamics complexity—delivered 2.2–$8.3\times$ on identical code and metrics.

The constructive reading: to help a latent world model, an inductive bias must supply what the latent *lacks*. Its state content is not what it lacks; its long-horizon dynamics are. Injections that target dynamics or conserved quantities rather than state, and extrinsic designs paired with OOD-robust encoders, are the two routes our results leave open—the former untested here beyond the suggestive velocity-on-parabola signature, the latter necessary but demonstrated insufficient alone. We note that our own evolution-targeting variants (the kinematic heads, the consistency loss, and the label-free prior) are deliberately simple instantiations chosen to span the design axes; their failure shows that pinning the evolution is not *automatically* useful, but it does not exhaust what a stronger, dynamics-matched evolution injection could achieve—designing one is future work.

Limitations. Our training-dynamics conclusions rest primarily on one compact backbone ($\sim$10M ViT-tiny JEPA); a frozen-encoder study across seven encoders up to 749M parameters supports the representational conclusions, and the headline findings—presence, the bypass, injection futility, and the protocol lever—replicate on a frozen-DINOv2 backbone (§4.5), where the sole apparent injection gain dissolves under content-free controls (Trap 5). On that backbone injection is inert rather than harmful, so the harm claim is established on the trainable-encoder regime. Headline comparisons, the re-weighting result, and the seed-refuted cells are three-seed; the remaining 26 scan cells are single-seed but sit 5–20 baseline seed-std above baseline. The Physion++ injection variants are single-seed with 3–$10\times$ gaps, and the probe family remains untested there. Physion and Physion++ are realistic *simulation*, not camera video. The one favorable cell (velocity on parabola) suggests dynamics-matched injection deserves systematic study; we did not perform it, and our evolution-targeting mechanisms are simple instantiations that may understate what a better-designed dynamics injection could deliver.

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder calibration controls.

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A Training and Evaluation Details

Hyperparameters. All PhyWorld runs: AdamW, lr $5{\times}10^{-5}$ (from-scratch reruns: $2{\times}10^{-5}$), weight decay 10^{-3}, bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime $2{\times}2$ uses 60 epochs uniformly), batch size 64 for free rollout (H=8), 128 for teacher forcing (H=1), 48 for $H{\geq}16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in[0.7, 1.5]$, velocity $\in[1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6]\cup[1.5, 2.0]$ and velocity from $[0, 0.8]\cup[4.5, 6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq$1,600 trajectories per domain).

Probe protocol. Ridge regression (α=1), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and K=4 stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads ρ=0.166 for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix E). Random-encoder and pixel-statistics controls are run alongside every probe.

B Full Injection Results

Table 5 is the numeric form of Figure 2, with the seed annotations spelled out.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.131	0.127	0.393
Fixed slot	0.183	0.156	0.651
+reweight (w=30)	0.114^{s}	0.160	0.596
+velocity	0.207	$\mathbf{0.096}^{\mathrm{m}}$	0.621
Probe	0.167	0.115^{n}	0.647
+slot+reweight	0.125^{s}	0.127	0.607
Kinematic (free MLP)	0.155	0.178	0.560
Kinematic (strict a=g)	0.206	0.173	0.559
Consistency	0.151	0.147	0.57–0.64
Label-free prior	0.171	0.172	0.653
Grounded prior	0.166	0.156	0.525

Table 5: The 30-cell scan (hardest clean split, latent nMSE $\downarrow$). $^{\mathrm{m}}$The one seed-robust gain: 0.096 ± 0.007 over three seeds vs. baseline $0.122{\pm}0.007$, non-overlapping. $^{\mathrm{n}}$Refuted by seeds: three-seed 0.137 ± 0.034 vs. baseline 0.122 ± 0.007—the single-seed low was seed luck. $^{\mathrm{s}}$Also refuted by seeds: reweighted slot 0.132 ± 0.017 and probe+slot $0.142{\pm}0.017$ vs. baseline $0.136{\pm}0.008$ (parity). Remaining cells are single-seed but sit 5–20 seed-std above baseline.

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the w=30 cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008. Kinematic head on the reweighted slot: velocity-OOD decoded position ρ=0.991/0.987 (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

Bypass probe details. Decoded position ρ per domain (uniform/parabola/collision). Baseline model: full 192 dims 0.92/0.85/0.78; black-box [2:192] alone 0.92/0.85/0.79. Slot-pinned model (structpos, w=30): slot dims [0:2] alone 0.92/0.85/0.51; black-box [2:192] alone 0.92/0.86/0.79—unchanged by pinning. The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$) without displacing the black box's redundant copy.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot's effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOD 0.131, parabola r/m-OOD 0.127, collision both-OOD 0.393):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity (+0.075/−0.064/+0.025) tracks velocity's role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection variants degrade the rigid-body scenarios by 3–10× (Table 3) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline (H=8) 0.016/0.140/0.280; H=20 0.022/0.033/0.220; H=16+scale 0.010/0.035/0.136; H=20+scale 0.012/0.024/0.087; H=28+scale reaches horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it,

H=20 explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.4) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds ($\sim$130 runs). Free-rollout baselines (three-seed): uniform both-OOD 0.427 ± 0.039, parabola r/m-OOD 0.225 ± 0.019, collision both-OOD 0.479 ± 0.009; Physion++ horizon-64 0.258 ± 0.005 vs. teacher forcing 1.030 ± 0.028.

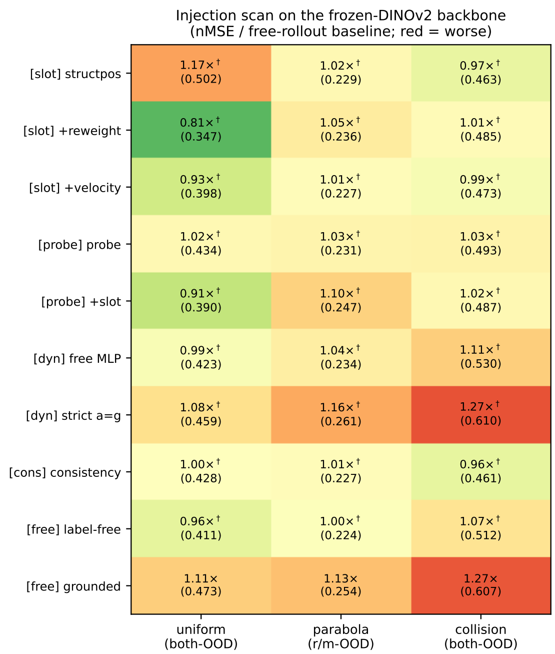

Figure 7: The injection scan repeated on the frozen-DINOv2 backbone (nMSE over the free-rollout baseline; mean over available seeds, † = three seeds). 10 cells worse / 17 parity / 3 better (27/30 at or below baseline). The dynamics-head variants are the worst here too (strict a=g and the grounded prior reach 1.27× on collision); the three above-parity cells, all on uniform, are the re-weighting confound of Trap 5.

Presence and bypass. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at ρ=0.951 (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the bypass are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, the core variants (re-weighted slot, probe, consistency) land within the baseline's seed band on all three domains, while the plain slot (1.42× on uniform) and the dynamics-head variants (1.15–1.29× on all domains) still hurt. With the encoder frozen, injection losses can only reshape the 384→192 adapter: the bypass cannot be displaced, which bounds the damage—and forecloses the benefit. Injection-variant variance also roughly doubles (±0.087 vs. baseline ±0.039 on uniform), so single-seed cells on this backbone are uninformative.

D Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper's main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform $0.131 \to 0.068$ (-48%), parabola -63% (r/m-OOD $0.127 \to 0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record ($0.393 \to 0.172 \pm 0.004$ over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062 \to 6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

E Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 distill the following five failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from ρ=0.44 to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to 100× (deform_clothhit: cosine $0.610 \to 0.913$, nMSE

$0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K{=}4$ probe ρ, $\lambda{=}50$ "won"; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes—$3{\times}10^4$–$2{\times}10^6$ at horizon ~ 28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine underreports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support ($+0.10$) and Collide ($+0.07$) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pretrained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2{=}0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from $0.427{\pm}0.039$ to $0.286{\pm}0.015$—apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches $0.330{\pm}0.013$, and weakening the pin $10\times$ while keeping the weighting reaches $0.329{\pm}0.084$—most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$–0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3, Test 2). At probe weight $\lambda{=}10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder's effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At $\lambda{=}1$ the ratios are benign, which is the setting used in §4.2.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.